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ZOO-506 Final Term Subjective

1) Scientific name of early shoot borer? 2

Trichogramma chilonis is the scientific name of early shoot borer.

2) Define arthropods? 2

A group with remarkable differences of structure and so abundant that all the other animals taken together are less than one-sixth as many as the arthropods

3) What is biological control? 2

It is possible, however, for man to encourage such beneficial forms by introducing them into new areas, by growing them artificially for liberation whenever they become scarce in nature.

4) Order of termites 2

Isoptera is the order of termites.

5) Appearance of adult rice-ear.2

The adult of Leptocorisa oratorius is green, light brown or mixed yellow in color. The adult are slender and about 20mm long.

6) Economically importance of insects. 2

No crop of either vegetable or animal origin appears to be entirely free from insect injury. Field, truck and fruit of crops, both growing and in storage, household goods and food products, forests and the wood products derived from them, domestic animals and animal products, all are more or less liable to insect attack.

7) Why fuzz is used? 2

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It is used in production of mattresses, surgical cotton, photographic film and paper.

8) What is sanitary control? 3

Methods to avoid introducing a pest into a field by

1- Cleaning field equipmentυ

2- Planting certified seedυ

3- Quarantinesυ

9) What are attractant? 3

Sometimes called attrahents are materials used to lure insects to traps and poisoned baits. The use of geraniol in Japanese beetle traps, sugar sirups for ant baits, and fermenting sirup solutions for trapping various moths are good examples.

10) Advantage of weeds destruction in cultural control? 3

The destruction of vegetable trash and weeds, frequently called clean culture, is also an important factor, especially in the control of hibernating insects. Weeds not only interfere with successful crop growth by crowding, but they may reduce the vigor of the crop by competing for plant food in the soil.

The destruction of weeds and brush along fences is of advantage in the reduction of insects.

But it results in the elimination of nesting and hiding places for insectivorous and game birds.

11) Why foreign insects brought in the country.3

From abroad to combat those foreign insect pests brought to this country accidentally but without their natural enemies.

12) Damage control by termites. 3

The use of partially decomposed manure should be avoidedυ Locate and destroy the termite colony nearby field.

Irrigation water with crude oil emulsion.

The removal of decaying organic matters cow dung, wood or dry stubbles from the field.

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Setts treatment with Imidacloprid (0.1%) or Chlorpyriphos 20 EC 0.04 % for 5υ min.

Application of well rotten Neem cake manure @ 60 Cartloads/ ha.υ Spray Chlorpyriphos 20%EC 750 ml/ha.

13) Time of planting of sugarcane for cultural control? 5

Time of planting:

1. Early planting during December –January to escape the shoot borer incidence.
2. Detrashing
3. Internode borer and Stalk borer
4. Manuring
5. Avoid excess use nitrogenus fertilizers.
6. Stalk borer and internode borer.
7. Trash mulching and earthing up.

14) Economic importance of insects? 5

No crop of either vegetable or animal origin appears to be entirely free from insect injury. Field, truck and fruit of crops, both growing and in storage, household goods and food products, forests and the wood products derived from them, domestic animals and animal products, all are more or less liable to insect attack. Although, it is the general practice to express losses in terms of dollar valuation. It probably would be safer to give them in bushels, tons, because the relative values of different crops and products change from year to year. Because there never has been a season free from the ravages of insects which might serve as a standard for comparison, much of this loss is not appreciated. If there could once be such a year entirely insect free, the difference would be apparent at once.

15) What things keep in mind during field scouting? 5

A field scout's objective is to provide an accurate and unbiased objective of pest populations.

What tools and procedures do you use for alfalfa weevil for potato leafhopper, etc?

Both of these examples are pests on alfalfa, however, the sampling procedure is different.

We cannot monitor their damage, nor are they easily counted on alfalfa foliage because they are small and easily fly when disturbed.

Instead, we must use and insect sweep net and count the number ofυ adults and nymphs caught.

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For second generation corn borers one sweet corn, we count the number of eggs laid on the leaves.

Different pests require different scouting practices.

16) Method of biological control. 5

Parasitic and predaceous insects as well as diseases of insects have already been discussed. As vital factors in the natural control of injurious insects. It is possible, however, for man to encourage such beneficial forms by introducing them into new areas. By growing them artificially for liberation whenever they become scarce in nature. Following this successful attempt many other parasites and predators have been imported.

Which in many instances have proved valuable in reducing the numbersυ of foreign pests.

17) Order, family and symptoms of tobacco cutworm. 5

Family: Noctuidae

Order: Lepidoptera.

Symptoms:

Scrapping the epidermal layer, leaving the skeleton of veins of leaf.

During severe attack, only the stem and side shoots will be standing in the field without any leaf or bolls.

Larvae feed the leaves by making small holes.

18) How shot and top and internode borer can be biologically controlled? 10

Cultivation of resistance variety

Irrigation at closer intervals for managingυ

Early shoot borer

Practice deep harvesting to destroy stubbles

Removal of water shoots to destroyυ

Top shoot borer

19) Pests of rice 10

Rice (Oryza sativa L.) is the most important staple food crop of the world

More than 60 % of world's population depend on it for food.υ

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1-Brown Plant Hopper, Nilaparvata lugens (Homoptera: Delphacidae)

2-Yellow Rice Stem Borer, Scirpophaga incertulas (Lepidoptera: Pyralidae)

3-Rice Ear-head Bug, Leptocorisa oratorius (Hemiptera: Alydidae)

4-Rice Hispa, Dicladispa armigera (Coleoptera: Chrysomelidae)

5-Rice Grasshopper, Hieroglyphus banian (Orthoptera: Acrididae)

20) Order coeloptera with example. 10

Includes: Beetles and Weevils, Blister beetle, Boll weevil, Collops beetle, Darkling beetle, Dermestid beetle, Dung beetle, Fig beetle, Flea beetle, Lady beetle, Long-horned beetle, Palo Verde root borer, Metallic wood borer.

Mouthparts: Chewing, Weevil- Piercing Sucking.

Metamorphosis: Complete Metamophosis.

21) Topbaror,internode baror,10

Internode borer

Constricted and shortened internodes, with a number of boreholes filled with excreta at nodal region Reddening of affected tissues inside the caneυ Fresh borer attack is mostly found in the top five immature internodes andυ its activity continues till harvest.

Top borer

Parallel rows of short holes in the emerging leaves causes a white streak which later turns reddish brown Dead heart in grown up canes reddish brown in color which cannot beυ easily pulled In tillering phase of the crop, the attacked shoots die, side shoots (tillers)υ develop Parallel rows of short producing a bunchy top appearance.

22) Appearance of stemborer 3

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The yellow stem borer is a specific pest of rice. The caterpillar alone are destructive, when full grown, they measure about 20mm and are dirty white or greenish yellow having brown head and pronotum.

23) Pink boll worm 2

Family: Gelechiidae

Order: Lepidoptera

It is a notorious pest of cotton in all cotton growing areas. It is a major chewing insect of cotton crop.

24) What is repellents and describe some repellents. 3

Repellents are those materials used to keep insects away from crops, animals and man. Various coal-tar and pine products are rather general insect repellents.

Naphthalene is a clothes moth repellent and oil of citronella is a common ingredient of preparations used to keep mosquitoes away from people.

25) Example of moisture manipulation to control insect. 2

A conspicuous example of the manipulation of moisture to control insects. Is the drying of grain to reduce the likelihood of weevil injury.

26) Low temperature in artificial control 5

The use of low temperatures in the artificial control of insects, therefore, is of two types. The more important is cold storage, as for furs, at temperatures just low enough to protect the stored materials from injury. The other use of cold actually to kill insects is feasible only on a large scale in zero weather in the north. Here as in the case of the heat-treatment much more time is required to chill grain, flour, clothing and other products than is usually realized.

27) High temperature in artificial control 3

High temperature is frequently used for killing insect infestations in soil and bulbs. Small lots of seeds and cereal products, Flour and feed sacks and Furniture, bedding and clothingυ

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28) Ear rice damages 5

The pest is essentially diurnal with highest activities in the morning and in the evening, seeking shelter during hotter parts of day.

Many generations are completed in a year.

Sucking of the grain sap causes empty or partially filled and chaffy grains and enhances subsequent fungal and bacterial infection.

Adult and nymph both suck the sap of developing rice grains at theυ milking stage and cause considerable yield loss.

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